Math Notes

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Contents

1 Statistics

1.1 Probability Space

In probability theory, a probability space or a probability triple $(\Omega, \mathcal{F}, P)$ is a mathematical construct that provides a formal model of a random process or "experiment".

Ω: A sample space, which is the set of all possible outcomes of a random process under consideration.
$\mathcal{F}$: An event space, which is a set of events, where an event is a subset of outcomes in the sample space. The σ-algebra $\mathcal{F}$ is a collection of all the events we would like to consider.
P: A probability function, which assigns, to each event in the event space, a probability, which is a number between 0 and 1 (inclusive).

In the example of the throw of a standard die:

The sample space Ω is typically the set $\{1, 2, 3, 4, 5, 6\}$.
The event space $\mathcal{F}$ could be the set of all subsets of the sample space, which would then contain simple events such as $\{5\}$ ("the die lands on 5"), as well as complex events such as $\{2, 4, 6\}$ ("the die lands on an even number"). So $\mathcal{F} = \{\{5\}, \{2, 4, 6\}, \dots\}$
The probability function P would then map each event to the number of outcomes in that event divided by 6. So for example: $P(\{5\}) = \frac{1}{6}$, $P(\{2, 4, 6\}) = \frac{3}{6}$, ...

1.2 Discrete Probability Distribution

Let X be a random variable with a finite number of finite outcomes $x_1, x_2, \dots, x_n$ occurring with probabilities $p_1, p_2, \dots, p_n$ respectively. The expectation of X is defined as

$$\mu = E[X] = \sum_{i=1}^{n} x_i p_i = x_1 p_1 + x_2 p_2 + \cdots + x_n p_n$$

The variance of X is

$$\sigma^2 = Var(X) = \sum_{i=1}^{n} p_i \cdot (x_i - \mu)^2$$

For a collection of n equally likely values:

$$\mu = \frac{1}{n} \sum_{i=1}^{n} x_i$$

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^{n} (x_i - \mu)^2$$

Expectation:
If X is independent of $\mathcal{H}$, then $E(X \mid \mathcal{H}) = E(X)$.
Tower property: for sub-σ-algebras $\mathcal{H}_1 \subseteq \mathcal{H}_2 \subseteq \mathcal{F}$, then $E(E(X \mid \mathcal{H}_2) \mid \mathcal{H}_1) = E(X \mid \mathcal{H}_1)$.
Law of total expectation: $E(E(X \mid \mathcal{H})) = E(X)$. A special case of tower property when $\mathcal{H}_1 = \{\emptyset, \Omega\}$.
Linearity: $E(X_1 + X_2 \mid \mathcal{H}) = E(X_1 \mid \mathcal{H}) + E(X_2 \mid \mathcal{H})$ and, $E(aX \mid \mathcal{H}) = aE(X \mid \mathcal{H})$ when $a \in \mathbb{R}$.

1.3 Continuous Probability Distribution

If the random variable X has a probability density function $f(x)$, and $F(x)$ is the corresponding cumulative distribution function, then

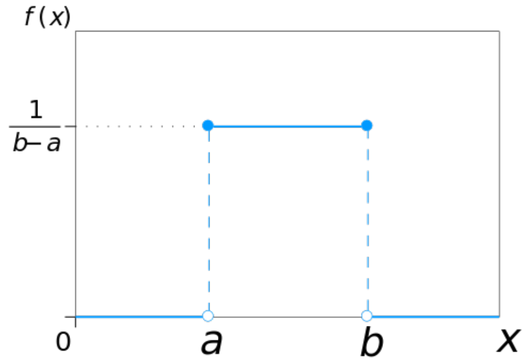
$$P(a \leq X \leq b) = \int_{a}^{b} f(x) dx$$

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)dt$$

$$\mu = \int_{\mathbb{R}} xf(x)dx = \int_{\mathbb{R}} xdF(x)$$

$$\sigma^2 = \int_{\mathbb{R}} (x-\mu)^2 f(x)dx$$

1.3.1 Uniform Distribution

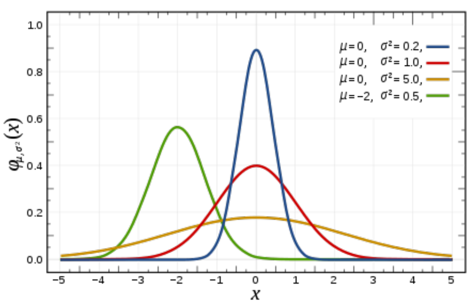


$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b, \\ 0 & \text{for } x < a \text{ or } x > b \end{cases}$$

$$\mu = \tfrac{1}{2}(a+b)$$

$$\sigma^2 = \tfrac{1}{12}(b-a)^2$$

1.3.2 Normal Distribution



$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2}$$

1.4 Law of large numbers

In probability theory, the law of large numbers is a mathematical law that states that the average of the results obtained from a large number of independent random samples converges to the true value, if it exists. More formally, the law of large numbers states that given a sample of independent and identically distributed values, the sample mean converges to the true mean.

2 Mathematical Analysis

2.1 Metric Space

A metric space is an ordered pair (M, d) where M is a set and d is a distance metric (function) on M,

$$d : M \times M \to \mathbb{R}$$

satisfying the following axioms for all points $x, y, z \in M$:

- The distance from a point to itself is zero: $d(x, x) = 0$
- (Positivity) The distance between two distinct points is always positive: If $x \neq y$, then $d(x, y) > 0$
- (Symmetry) The distance from x to y is always the same as the distance from y to x: $d(x, y) = d(y, x)$
- The triangle inequality holds: $d(x, z) \leq d(x, y) + d(y, z)$

Cauchy Sequence
Given a metric space (M, d), a sequence of elements of X, $x_1, x_2, x_3, \ldots$ is Cauchy, if for every positive real number $\epsilon > 0$ there is a positive integer N such that for all positive integers $m, n > N$, the distance $d(x_m, x_n) < \epsilon$.

Complete Metric Space
In mathematical analysis, a metric space M is called complete (or a Cauchy space) if every Cauchy sequence of points in M has a limit that is also in M.

2.1.1 Contraction Mapping

In mathematics, a contraction mapping, or contraction or contractor, on a metric space (M, d) is a function f from M to itself, aka contraction mapping $f : M \to M$, with the property that there is some real number $0 \leq k < 1$ such that for all x and y in M,

$$d(f(x), f(y)) \leq k d(x, y)$$

2.1.2 Banach fixed-point theorem

Let (X, d) be a non-empty complete metric space with a contraction mapping $T : X \to X$. Then T admits a unique fixed-point x^* in X, $T(x^*) = x^*$. Furthermore, x^* can be found as follows: start with an arbitrary element $x_0 \in X$ and define a sequence $(x_n)_{n \in \mathbb{N}}$ by $x_n = T(x_{n-1})$ for $n \geq 1$. Then $\lim_{n \to \infty} x_n = x^*$. (Wikipedia proof)